ANALYSIS OF RATE LIMITING STRATEGIES IN WEB SYSTEMS

PROBLEM

- Too many requests → system overload
- No control → instability
- Need request management

SOLUTION

- Too many requests → system overload
- No control → instability
- Need request management

IMPLEMENTED STRATEGIES

- No Rate Limiting
- Fixed Window
- Sliding Window

HOW SYSTEM WORKS

Client → Request → Rate Limiter → Accept / Reject

CODE LOGIC

- Check request count
- Compare with limit
- Accept or reject

EXPERIMENT SETUP

- Multiple clients
- High request rate
- Different strategies

RESULTS

- No limit → high throughput
- Fixed window → burst behavior
- Sliding window → stable

ANALYSIS

- Trade-off: performance vs protection
- Different behaviors under load

TECHNICAL INSIGHT

- Fixed window problem: burst
- Sliding window: precise control

CONCLUSION

- Rate limiting is essential
- Strategy matters

THANK YOU

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